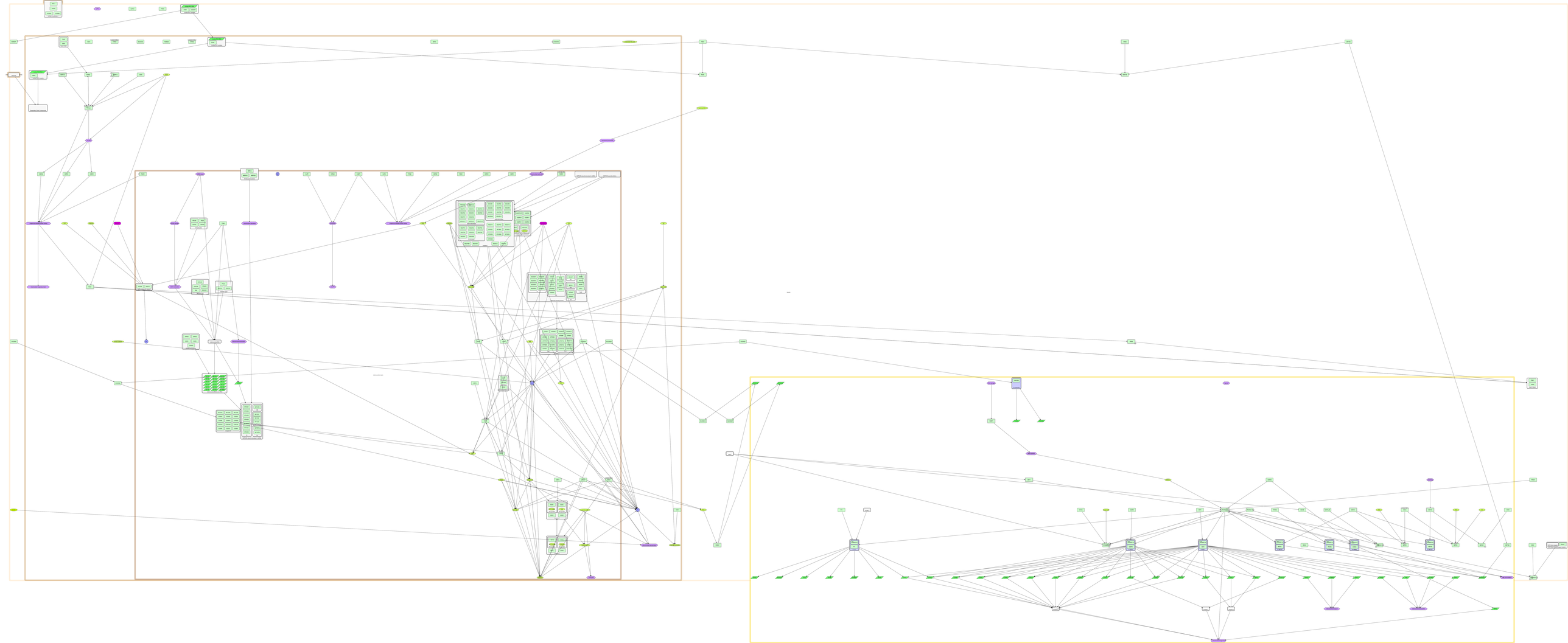

SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
UPS
NDN
PPARGC1A
AMPK
CAPN1
PROX1
PINK1
PRKN
ZNF746
UPS
nucleus
PARP1
DNA damage
SIRT1
SNCA
ZNF746
PARP1
NAD depletion
NAD(+)
PPARGC1A
GSK3B
PPARGC1A
PPARGC1A
S538
T177
acetyl-CoA
MAPK p38
KAT2A
RNS
NO
SNO
MEF2C
MEF2C
mutated A53T
SNCA
ROS
SOH
MEF2C
MEF2D
autophagy
CDK5
MEF2D
PPARA
ESRRA
YY1
GABPA
NRF1
mTORC1
mitochondrial biogenesis
T295
PPARGC1A
PPARA
TF PPARA
ESRRA
TF ERRa
YY1
TF YY1
GABPA
TF NRF2
NRF1
TF NRF1
MEF2D
TF MEF2D
MEF2C
TF MEF2C
PPARGC1A
fatty acid oxidation
SIRT3
UCP2
IDH3A
IDH3B
IDH3G
NDUFS8
ATP5MC1
COX5A
CYCS
COX4I1
COX5A
COX5B
COX6A1
COX6B1
COX6C
COX7A2
COX7B
COX7C
COX8A
CYCS
SDHB
SDHC
SDHD
ATP5F1B
ATP5F1C
POLRMT
TFAM
TFB2M
TFB1M
mitochondrial transcription
SURF1
VDAC1
TOMM20
mitochondrial transport
complex IV
complex II
complex V
mitochondrial biogenesis
CHCHD2
TF CHCHD2
COX4I2
CHCHD2
hypoxia
ROS
CHCHD4
CHCHD2
SLC25A27
SLC25A14
SLC25A27
SLC25A14
ROS
PARK7
SLC25A27
SLC25A14
CCCP
peroxynitrite
SIRT3
NDUFA9
NDUFA10
PINK1
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
O2
superoxide
hydrogen peroxide
CHCHD2
H+
SOD1
mitochondrial depolarization
membrane permeability
superoxide
purine nucleotide
unsaturated fatty acid
Tiopronin
diazoxide
ATP
UQCRH
UQCRFS1P1
UQCRQ
ubiquinone
mutated A53T
NADH
NAD(+)
rotenone
iron-sulfur cluster
iron-sulfur subunit
heme b
MT-CYB
MT-CYB:heme b
complex III
CHCHD2:CHCHD4:CYCS
CYCS
MT-CO1
MT-CO2
MT-CO3
COX7A1
COX7B
COX7C
COX8A
complex IV
O2
H2O
ATP5F1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5PD
ATP5MG
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
ADP
ATP-sensitive K channel
KCNJ8
KCNJ11
K+
ROS
Ca2+
mitochondrial dysfunction
SIRT5
myxothiazol
SLC25A27
SLC25A14
UCP2
UCP3
superoxide
OXPHOS assembly factors
O2
superoxide
O2
superoxide
SDHB
SDHA
Fe-S cluster
FAD
iron sulfur protein
flavoprotein
SDHC
SDHD
complex II
SIRT3
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
OXPHOS assembly factors
SQSTM1
PARK7
OXPHOS subunits encoded in mtDNA
ubiquinol
succinic acid
TCA cycle
ubiquinone
fumaric acid
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
CI
MT-CYB
CIII
MT-CO1
MT-CO2
MT-CO3
CIV
MT-ATP6
MT-ATP8
CV
OXPHOS subunits encoded in mtDNA
MT-TC
MT-TL1
MT-TS1
MT-TE
MT-TL2
MT-TS2
MT-TF
MT-TN
MT-TT
MT-TH
MT-TP
MT-TV
MT-TI
MT-TQ
MT-TW
MT-TK
MT-TR
MT-RNR1
mitochondrial tRNAs and rRNA
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
mRNA
mitochondrial translation
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
mitochondrial transcription
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
mitochondrial DNA
mtDNA repair
TFAM
POLG
ERCC6
ERCC8
Mt-DNA repair
nucleoid
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
mitochondrial matrix
mitochondrion
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
peroxisome
CDK5
PARL
HAX1
HAX1:PARL
mutated A141S
HTRA2
mutated G399S
HTRA2
Y182
T180
MAPK14
MAPK12
S400
HTRA2
HTRA2
HTRA2
HAX1
ROS
PINK1
TRAP1
PINK1
TRAP1
mt UPR
HTRA2
LONP1
CLPB
HSPD1
HSPE1
HSPA9
HSPD1
HSPE1
HSPA9
CLPP
chaperone-mediated protein folding
chaperone-mediated protein folding
proteolysis
mt UPR
PINK1
TFAM
mitochondrial respiratory chain
PINK1
PRKN
PINK1
PRKN
PINK1:PRKN
Respiratory Chain Components
ribosome
mitochondrion-attached ribosome
ribosome
TOMM20
PINK1
nuclear RCC RNA
PINK1
nuclear RCC complex
nuclear RCC RNA
PUM1
nuclear RCC complex
TOMM20
PINK1
PUM1
&
nuclear RCC RNA
PUM1
nuclear RCC complex
PINK1
HNRNPF
&
PRKN
nuclear RCC RNA
PUM1
HNRNPF
nuclear RCC complex
neuron

MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
UPS
CAPN1
PRKN
nuclear RCC RNA
PUM1
HNRNPF
nuclear RCC complex
HNRNPF
PARL
HAX1
HAX1:PARL
HAX1
HTRA2
NDUFA10
TOMM20
HTRA2
SIRT3
NDUFA9
unsaturated fatty acid
PINK1
PUM1
PRKN
ZNF746
ZNF746
ribosome
Respiratory Chain Components
MAPK12
HTRA2
MAPK14
CDK5
ROS
mt UPR
HSPA9
HSPE1
HSPD1
peroxynitrite
membrane permeability
TRAP1
mtDNA repair
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
Ca2+
CLPP
HTRA2
LONP1
CLPB
TFAM
HSPA9
PINK1
HSPD1
HSPE1
mitochondrial dysfunction
PARK7
OXPHOS subunits encoded in mtDNA
OXPHOS assembly factors
chaperone-mediated protein folding
ATP
Tropronin
mtDNA damage
POLG2
POLG
MTFMT
MTPAP
Mt replication
TFAM
mitochondrial translation
proteolysis
chaperone-mediated protein folding
ROS
rotenone
O2
O2
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
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NDUFC2
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UQCRFS1P1
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UQCR11
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UQCRC2
MT-CYB
heme b
MT-CYB:heme b
complex III
mitochondrial respiratory chain
PINK1
KCNJ8
KCNJ11
ATP-sensitive K+ channel
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
POLG
ERCC6
ERCC8
Mt-DNA repair
mt UPR
superoxide
NDUFAF1
NDUFAF2
NDUFAF3
NDUFAF4
NDUFAF5
NDUFAF6
ACAD9
FOXRED1
CI
COX10
COX14
COX15
FASTKD2
LRPPRC
SCO1
SCO2
SURF1
TACO1
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
neuron
CHCHD4
purine nucleotide
K+
AARS2
DARS2
GARS
KARS
RARS2
Mt-tRNA synthetases
mitochondrial DNA
mitochondrial transcription
UCP2
UCP3
ADP
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MF
ATP5PO
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP8
ATP5PF
MT-ATP6
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
SLC25A14
SLC25A27
CHCHD2
mitochondrial matrix
CHCHD2
MT-TC
MT-TL1
MT-RS1
MT-TE
MT-TL2
MT-TS2
MT-TF
MT-TN
MT-TT
MT-TH
MT-TP
MT-TV
MT-TI
MT-TQ
MT-TW
MT-TK
MT-TR
MT-RNR1
mitochondrial tRNAs and rRNA
mRNA
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
CI
MT-CYB
CIII
MT-CO1
MT-CO2
MT-CO3
MT-ATP6
MT-ATP8
CV
OXPHOS subunits encoded in mtDNA
SIRT5
GHITM
CHCHD2
CYCS
CHCHD2:GHITM:CYCS
H+
ATP
CYCS
H2O
CYCS
NADH
ubiquinone
SIRT3
SNCA
SNCA
ROS
NAD(+)
SDHB
Fe-S cluster
Fe-S protein
SDHA
FAD
flavoprotein
SDHC
SDHD
complex II
succinic acid
H+
SOD1
SDHB
Fe-S cluster
Fe-S protein
SDHA
FADH2
flavoprotein
SDHC
SDHD
complex II
fumaric acid
mitochondrial depolarization
hydrogen peroxide
PARK7
CCCP
ubiquinol
TCA cycle
SLC25A14
SLC25A27
SLC25A14
SLC25A27
DNA damage
CHCHD2
TF CHCHD2
PARP1
COX4I2
CHCHD2
hypoxia
NAD depletion
AMPK
SIRT1
NAD(+)
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autophagy
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PINK1:PRKN
YY1
mTORC1
KAT2A
acetyl-CoA
GABPA
NRF1
PPARGC1A
PPARGC1A
PPARA
GSK3B
MAPK p38
MEF2C
ROS
SNCA
MEF2D
RNS
NO
CDK5
PPARGC1A
YY1
TF YY1
PPARGC1A
PPARGC1A
GABPA
TF NRF2
PPARGC1A
NRF1
TF NRF1
PPARGC1A
MEF2C
TF MEF2C
SNCA
PPARGC1A
PPARA
TF PPARA
PPARGC1A
ESRRA
TF ESRRA
PPARGC1A
MEF2C
PPARGC1A
MEF2D
TF MEF2D
MEF2C
MEF2D
ZNF746
NDN
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
PPARGC1A
IDH3G
NDUFS8
UCP2
IDH3B
ATP5MC1
IDH3A
CYCS
COX5A
COX5B
COX4I1
COX6B1
COX7B
COX6C
COX7B
COX6C
COX6A1
COX7C
ATP5F1B
COX8A
COX7A2
SDHB
SDHD
SDHC
ATP5F1C
SURF1
TOMM20
VDAC1
TFB2M
TFAM
POLRMT
TFB1M
PPARGC1A
fatty acid oxidation
complex IV
complex IV
complex V
complex II
mitochondrial transport
mitochondrial transcription
SIRT3
mitochondrial biogenesis